



Designing for the unknown

While we can not predict every interference scenario, we can improve our design resilience:

- **Choose your frequency wisely.** Avoid crowded ISM bands when possible. Lower-frequency or less congested bands may offer better reliability.
- **Use hardware filters.** A SAW filter can simultaneously reduce emissions and improve out-of-band immunity.
- **Consider shielding.** RF shielding, ferrite beads, and robust PCB layout practices can make a massive difference in noisy environments.
- **Design for security, not just functionality.** Build secure stacks, validate inputs, isolate systems, and avoid assumptions about trusted environments.

heavily influenced by the surrounding environment.

Designing with margins and fall-back mechanisms

In wireless design, margins must be reconsidered compared to wired systems. For BLE, you need a minimum RSSI of -80dBm for a stable connection, but design for -50dBm, accounting for a 30dB margin due to unpredictable RF loss, obstructions, and reflections. This ‘worst-case design’ ensures system robustness in variable environments.

Over-the-air (OTA) updates, which are crucial for vehicle systems such as bug fixes and security patches, introduce risks. A failed or buggy OTA update can disable BLE-dependent features, like TPMS or PaaS, while leaving Bluetooth audio unaffected. To mitigate this, implement fallback mechanisms, such as A/B partitioning, which allows the system to revert to the last stable configuration. OTA resiliency is now a requirement.

This case study emphasises the importance of designing for variability, anticipating potential failures, and incorporating generous margins to ensure reliable user experiences, from keyless entry to tire health monitoring.

Another vital aspect in building connected products that requires attention from design to post-production is cybersecurity. Security

and connectivity are intertwined—every wireless interface is a potential attack vector. As vehicles become increasingly connected, ensuring secure, authenticated, and encrypted communications is crucial for maintaining system reliability and integrity.

A 2015 Jeep hack exemplified the risks of connected vehicles. Hackers remotely took control of a Jeep Cherokee, shutting down the engine, proving that security vulnerabilities in connected systems are a serious concern. This event highlighted the importance of robust security in modern automotive design.

The modern car: A hacker’s playground

Today’s vehicles are not just machines—they are connected networks on wheels. With features like Wi-Fi, Bluetooth, cellular, and cloud synchronisation, modern cars are filled with communication channels, each offering potential vulnerabilities. A single weak link could give attackers access to the controller area network (CAN bus), the central system controlling everything from brakes to steering. It’s as if someone could take full control of your car without ever being nearby.

To tackle these risks, a secure system architecture is key: isolating communication stacks, hardening interfaces, validating inputs, and building security into the

design process from the start. This approach isn’t just important for cars; the medical field has seen similar cybersecurity failures. For instance, the FDA once recalled pacemakers, not due to faulty hardware, but because of severe cybersecurity flaws. These devices lacked authentication, meaning attackers could send commands or access sensitive data. Without encryption, attackers could drain the device’s battery by triggering unnecessary wakeups.

This was not an impossible engineering challenge; it was a basic oversight in security design. While the issue was later fixed via software, it highlights the critical importance of security in life-critical systems. It is a reminder that in today’s connected world, security isn’t a reactive fix but a proactive necessity.

Security: The elephant in the room

These stories—whether from highways, hospitals, or garages—highlight a simple but profound truth: the world is unpredictable. Attack vectors aren’t always obvious. Interference isn’t always explainable. However, the systems we build must withstand both.

Secure, resilient design isn’t a checkbox at the end of development. It is a philosophy that must be baked into the foundation, saving us not just from bad press and recalls, but from real-world consequences that could impact lives.

If there is one lesson we must carry forward, it’s this: connectivity is power, but only when paired with responsibility.

In today’s hyper-connected world, building a product no longer ends at getting the hardware right. The journey from a great piece of technology to a successful, user-friendly product is layered with